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Effects of a metabolizable protein supply on metabolic function of CD4⁺ T lymphocytes in mid-lactation dairy cows

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ABSTRACT

The objective of the present experiment was to determine the effects of MP supply on metabolic function of nonactivated (NA) and activated (AC) CD4⁺ T lymphocytes in mid-lactation dairy cows. Thirty-two multiparous Holstein cows with a mean $\pm$ SD lactation of 3.28 ± 0.99 , milk yield 51.6 ± 6.1 kg/d, and milk protein $3.38\% \pm 0.47\%$ were enrolled at 86 ± 18 DIM in a barn equipped with 16 bin feeders. Cows and bins (16 cows/8 bins per treatment) were randomly assigned to treatments. All cows were fed the same herd diet (MP = 3,006 g/d) during a 2-wk adaptation period before enrollment and a 1-wk covariate period following enrollment to allow acclimation to the bins and collect baseline measurements. Following the covariate period, cows received 1 of 2 treatments for 12 wk: diets formulated to be either adequate in MP supply (MPA; MP = 3,111 g/d) or MP inadequate (MPI; MP = 2,983 g/d). The difference in MP supply was achieved by including porcine blood meal in the MPA, whereas it was omitted in the MPI treatment. Blood was collected during wk 11 at 173 ± 18 DIM to isolate peripheral blood mononuclear cells followed by magnetic isolation of CD4⁺ T lymphocytes using bovine-specific monoclonal antibodies. Isolated CD4⁺ T lymphocytes from each cow were split into 2 tubes and randomly assigned to incubate in an assay medium (250 μ L) or with a combination of phorbol 12-myristate 13-acetate (100 μ L, 20 ng/mL) and ionomycin (10 μ L, 1 μ g/mL) for 2 h at 37 °C to evaluate metabolic function under metabolically NA and AC states, respectively, using an extracellular flux analyzer. Mitochondrial and glycolytic functional kinetics were recorded for NA and AC CD4⁺ T lymphocytes based on real-time measurement of oxygen consumption rate (OCR), extracellular acidification rate (ECAR), and proton efflux rate (PER) under basal condi-

tions and in response to complex V inhibitor (oligomycin A), a protonophore uncoupler (BAM 15), and complex I and complex III inhibitors (rotenone and antimycin A). Cows fed the MPI diet exhibited lower OCR leading to a reduced maximal respiration compared with MPA-fed cows. Feeding MPI diet decreased maximal respiration in both metabolically NA and AC CD4⁺ T lymphocytes. Feeding MPI diet decreased sparing respiratory capacity in both metabolically NA and AC CD4⁺ T lymphocytes. During the T-cell persistence assay, an interaction between diet and time within the assay was observed for ECAR and PER, indicating that CD4⁺ T cells from cows fed MPI showed a progressive decrease in aerobic glycolysis over the 1.50 to 73.0 min measurement period, whereas CD4⁺ T cells from cows fed MPA maintained higher glycolytic activity. Mitochondrial, glycolytic, and total ATP production rates were unaffected by MP supply; however, glycolytic ATP production rates were greater in AC than NA CD4⁺ T lymphocytes. These results indicate that suboptimal supply of MP reduced the metabolic fitness and persistence of CD4⁺ T lymphocytes, which might compromise immune responses during nutrient imbalance, such as early lactation or health challenges.

Key words: amino acids, dairy cow, metabolism, T lymphocytes

INTRODUCTION

Corn- and soybean-based diets commonly fed to dairy cows often result in deficiencies in EAA, particularly Lys and Met (Schwab and Broderick, 2017; NASEM, 2021). These EAA are critical not only for milk and milk component synthesis but also for supporting key metabolic and immune functions (Zhou et al., 2016; Schwab and Broderick, 2017; Batistel et al., 2018). To meet metabolizable AA requirements, diets are traditionally supplemented with high-AA feed ingredients such as blood meal or rumen-protected (RP) AA products. Supplementation of AA in a RP form, such as RP-Lys, RP-Met, or RP-His, to increase the supply of metabolizable EAA during the

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transition or lactation periods has been shown to enhance productive performance in dairy cows (Zanton et al., 2014; Räisänen et al., 2023; Arshad et al., 2024a). Notably, Westhoff et al. (2024) observed that increasing the total metabolizable AA supply from 1,175 to 1,603 g/d prepartum and from 2,044 to 2,685 g/d postpartum resulted in an ECM increase of 7.20 kg/d, despite similar supplies of metabolizable Met and Lys across treatments. These findings highlight the importance of considering not only the profile of individual AA but also the overall adequacy of total metabolizable AA supply in supporting optimal performance during mid lactation.

Beyond their traditional role in supporting milk or milk components synthesis, AA are also recognized as regulators of immune function, acting through roles in antioxidant defense, inflammatory modulation, and immune cell signaling (Pearce et al., 2013; O'Neill et al., 2016). At the cellular level, AA can directly influence T lymphocyte proliferation, cytokine production, and metabolic activity by serving as substrates for protein synthesis and signaling molecules that regulate mTOR and other nutrient-sensing pathways (Wang and Green, 2012; Sinclair et al., 2013). Evidence from nonruminant models indicates that restricting or supplementing these AA can rapidly alter T-cell bioenergetics and activation potential, highlighting a direct link between dietary AA availability and immune cell function. In fact, supplementation with RP-Met during the transition period has been shown to enhance neutrophil phagocytic and oxidative burst capacity, while concurrently promoting antioxidant defense mechanisms through elevated synthesis of glutathione and reducing circulating concentrations of inflammatory biomarkers such as haptoglobin (Zhou et al., 2016; Batistel et al., 2018). Parallel evidence indicates that RP-Lys supplementation during the transition period attenuates systemic inflammation, evidenced by reduced circulating haptoglobin concentrations and glutathione peroxidase activity in dairy cows (Fehlberg et al., 2023). These findings collectively suggest that the positive effects of AA supplementation on production performance may be mediated, at least in part, by nutrient partitioning that favors lactation and alleviates the energetic demands imposed by immune activation (Kvidera et al., 2017; Arshad et al., 2024b; Westhoff et al., 2025a).

Despite growing recognition of nutrient-immune interactions in ruminants, most studies have centered on systemic immune responses or broad leukocyte function, often within the context of transition-associated metabolic stress (Batistel et al., 2018; Fehlberg et al., 2023; Westhoff et al., 2025b). Immune activation is energetically demanding and requires metabolic reprogramming, including shifts between glycolysis and oxidative phosphorylation (Arshad et al., 2024b). Consistently, bovine T helper cells undergo metabolic rewiring during demand-

ing stages such as early lactation, enhancing glycolytic and oxidative capacity to support activation and effector function, which may reflect immune dysregulation rather than classical immunosuppression (Fernandez-Wallace et al., 2026). Nevertheless, the extent to which MP supply influences immune cell bioenergetics remains unclear in mid-lactation dairy cows. We focused on CD4⁺ T lymphocytes, which are central to adaptive immunity and highly sensitive to nutrient-dependent metabolic regulation (Pearce et al., 2013). Their rapid transition between catabolic and anabolic states during activation makes them a sensitive indicator of nutrient availability, particularly under marginal protein supply (Pearce et al., 2013; O'Neill et al., 2016; Bantug et al., 2018). Although innate lymphoid cells are also nutrient responsive, CD4⁺ T cells remain the most established mechanistic model linking metabolism to immune function (van der Windt and Pearce, 2012; Basso et al., 2024), making them an ideal starting point to test whether MP deficiency alters immunometabolic prioritization.

Given this rationale, we evaluated mitochondrial and glycolytic metabolism in CD4⁺ T lymphocytes isolated from high-producing mid-lactation dairy cows fed either an MP-adequate (MPA) or MP-inadequate (MPI) diet. Mid-lactation cows were chosen as an ideal model due to their physiologically stable state, which minimizes confounding factors from the inflammatory and metabolic disturbances typical of the periparturient period. Although T-cell activation is known to increase metabolic demands by promoting aerobic glycolysis and mitochondrial respiration in both ruminants and nonruminants (Pearce et al., 2013; Palsson-McDermott and O'Neill, 2013; Arshad et al., 2024b), it remains unclear whether dietary MP supply affects metabolic responses in dairy cows. We hypothesized that feeding an MPA diet would enhance the metabolic function of CD4⁺ T lymphocytes. The objective of this experiment was to evaluate the effects of MP supply on mitochondrial and glycolytic metabolism in both nonactivated (NA) and activated (AC) CD4⁺ T lymphocytes in high-producing mid-lactation dairy cows.

MATERIALS AND METHODS

The experiment was conducted at the University of Wisconsin–Madison Emmons Blaine Dairy Research Center (Arlington, WI), and all animal procedures were approved by the University of Wisconsin–Madison College of Agricultural and Life Sciences Animal Care and Use Committee, protocol number A006420. This article is one of 2 companion articles (Arshad et al., 2026) from an experiment designed to examine the effects of replacing porcine blood meal (PBM) with RP-Lys on produc-

tive performance, ECM:feed ratio, nutrient partitioning, and blood metabolites in dairy cows.

Sample Size Calculation

Sample size was calculated a priori using the POWER procedure of SAS (version 9.4, SAS/STAT, SAS Institute Inc., Cary, NC), accounting for the split-plot structure of the experimental design. Specifically, dietary treatments (MPA vs. MPI) were treated as whole-plot factors, whereas cell-level treatments (NA vs. AC) were treated as sub-plot factors within each animal. Power calculations were performed independently for the whole-plot and sub-plot comparisons to reflect the different levels of variability associated with each factor. For the whole-plot comparison (MPA vs. MPI), a mean difference of 24 pmol/min in sparing respiratory capacity (SRC) and a between-cow SD of 18 pmol/min were assumed based on previous findings (Arshad et al., 2024c). Under these assumptions, a minimum of 10 cows per dietary group was required to achieve 80% power at $\alpha = 0.05$ using a 2-tailed, 2-sample *t*-test. For the sub-plot comparison (NA vs. AC), where each cow served as its own control, a paired-sample power analysis was conducted. Based on a within-cow SD of 15 pmol/min, an expected mean difference of 13 pmol/min in SRC, and a correlation coefficient of 0.74 between paired measures (Arshad et al., 2024b), a minimum of 8 cows with both NA and AC measurements was required to achieve 80% power at $\alpha = 0.05$ using a 2-tailed paired *t*-test. Although the minimum required sample sizes were 10 cows per dietary group and 8 cows for within-cow comparisons, a total of 32 cows were enrolled. This was to ensure adequate statistical power for both levels of inference while accounting for potential dropouts due to clinical or subclinical health issues, which could confound systemic energy metabolism and T lymphocyte function.

Experimental Design, Dietary Treatments, Cows, and Housing

Multiparous Holstein cows ($n = 32$) with a mean $\pm$ SD lactation of 3.28 ± 0.99 , milk yield 51.6 ± 6.1 kg/d, and milk protein $3.38\% \pm 0.47\%$ were enrolled at 86 ± 18 DIM in this study. Cows were housed in a pen equipped with 32 electronic bin feeders (RIC system, Insentec, Marknesse, the Netherlands) at the University of Wisconsin–Madison Emmons Blaine Dairy Research Center in a sand-bedded freestall barn (Arlington, WI) for 15 wk. The electronic feeding system was described by Chapinal et al. (2007). Cows (experimental units) and bins (16 cows/8 bins per treatment) were randomly assigned across 2 treatments (MPA = 3,111 g/d vs. MPI = 2,983 g/d), and housing conditions were as described in

the companion article (Arshad et al., 2026). Gate feeders were divided into 4 groups of 8 (representing the first quarter, second quarter, third quarter, and fourth quarter of the pen), and treatments were randomly assigned to each group of 8 (2 treatments per quarter of the pen) to ensure that treatments were evenly distributed to gate feeders across the length of the pen. Before study enrollment, cows underwent a 2-wk adaptation period to acclimate to the bins, followed by a 1-wk covariate period during which baseline measurements of DMI and productive performance were recorded (Arshad et al., 2026; Figure 1A).

During both the adaptation and covariate periods, all cows received the same herd diet (MP = 3,006 g/d). Following the covariate period, 1 of 2 treatment diets (MPA vs. MPI) were fed for 12 wk. These treatments were part of a larger experiment (Arshad et al., 2026) in which PBM (Consumers Supply, Sioux City, IA; 16 h RUP 87.5%, crude fat 1.0%, crude fiber 2.0%; concentration of Lys = 8.14%) was progressively replaced with RP-Lys (LysiGEM Extend, Kemin Industries Inc.) across 4 dietary treatments. Three samples of PBM from the same batch used in the experiment were analyzed by Dairyland Laboratories Inc. (Arcadia, WI) to determine the undigestible CP (UCP) fraction. Undigestible CP represents the portion of CP resistant to both rumen microbial degradation and enzymatic digestion in the intestine, and was measured using *in vitro* techniques that simulate rumen conditions followed by enzymatic digestion. The mean ($\pm$ SD) value of UCP of PBM was 26.4 ± 1.0 . The original treatments in Arshad et al. (2026) included a basal diet with PBM as the primary source of metabolizable Lys, and 3 diets where 33%, 66%, or 100% of the metabolizable Lys from PBM was replaced with RP-Lys (LysiGEM Extend, Kemin Industries Inc.) as **RPL33**, **RPL66**, or **RPL100**. The supply of metabolizable Met was maintained similar across all 4 treatments by supplementing with RP-Met (KESSENT Me, Kemin Industries Inc.). The RP-Lys product (LysiGEM Extend, Kemin Industries Inc.) contained 60% L-Lys with 49% bioavailability (29.4 g of metabolizable Lys per 100 g of product). The bioavailability of Lys was determined using the free AA dose-response technique developed by Whitehouse et al. (2017), with this specific product evaluated in a subsequent study (Whitehouse et al., 2023). The RP-Met product (KESSENT Me, Kemin Industries Inc.) contained 74% DL-Met with 80% bioavailability (59 g of metabolizable Met per 100 g of product), determined using the seleno-Meth technique established by Weiss and St-Pierre (2009), with this product evaluated in Parales-Giron et al. (2022). In the present article, metabolic function of CD4⁺ T lymphocytes was evaluated in cows assigned to 2 of these treatments: the PBM-based diet (MPA) and the RPL100 diet (MPI). These diets

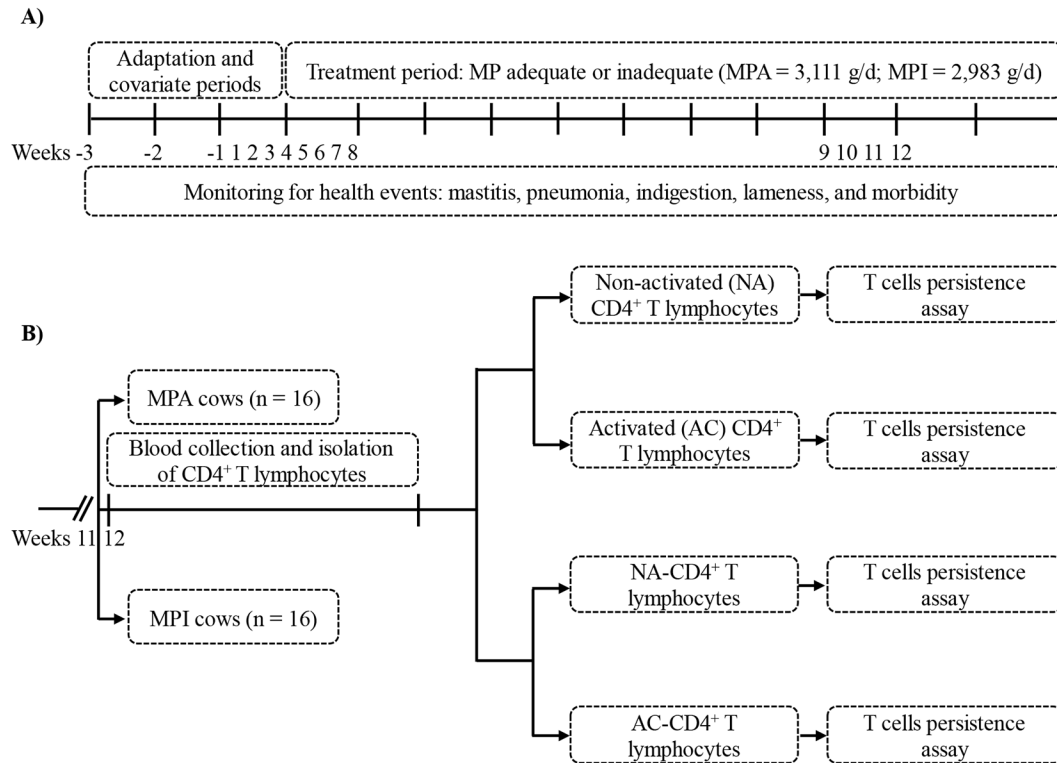


Figure 1. Schematic representation of the experimental design evaluating the effects of MP supply on the metabolic function of nonactivated (NA) and activated (AC) CD4⁺ T lymphocytes in mid-lactation dairy cows. Multiparous Holstein cows ($n = 32$) were enrolled at 86 ± 18 DIM. As illustrated in panel A, following a 2-wk adaptation period and a 1-wk covariate period, cows started receiving 1 of 2 treatments: diets formulated to be either adequate in MP supply (MPA; MP = 3,111 g/d) or inadequate (MPI; MP = 2,983 g/d) for 12 wk. Additional samples were collected during wk 11 and 12, as shown with more detail in panel B. Blood samples were collected during wk 11 (173 ± 18 DIM) for peripheral blood mononuclear cell isolation, followed by magnetic separation of CD4⁺ T lymphocytes. Isolated CD4⁺ T lymphocytes from each cow were incubated either with an assay medium to remain metabolically NA or AC using a combination of assay medium, phorbol myristate acetate, and ionomycin. A Seahorse-based T-cell persistence in vitro assay was performed to assess metabolic parameters, including the oxygen consumption rate (a measure of oxidative phosphorylation), as well as extracellular acidification and proton efflux rates (measures of cellular glycolysis), all measured in real time using an extracellular flux analyzer.

were selected to contrast adequacy versus deficiency in overall MP supply, while providing similar supplies of metabolizable Lys and Met. Diet composition and ingredient details are provided in Table 1, whereas the supply of EAA across both treatments, estimated according to NASEM Dairy (NASEM, 2021), is shown in Table 2.

Characterization of Health Events

Data on health events were retrieved retrospectively for the 13-wk study period from archived computer backup files of the on-farm dairy management software (DairyComp 305; Valley Agricultural Software). Clinical mastitis was diagnosed by herd personnel immediately before each milking based on abnormal milk appearance in one or more quarters and confirmed using the California Mastitis Test and bacteriological culture plates, following standard herd procedures. Pneumonia was identified in cows exhibiting a respiration rate greater than 60 breaths per minute, abnormal lung sounds detected

upon auscultation, and concurrent fever ($>39.5^{\circ}\text{C}$). Indigestion was diagnosed based on reduced rumen motility observed during auscultation and ballotement, a decline in feed intake (as-is basis) of $\geq 30\%$ relative to the individual's weekly average, and abnormal fecal consistency. Lameness was assessed using a standardized locomotion scoring system (Sprecher et al., 1997), in which cows scoring ≥ 3 on a 1 to 5 scale were classified as clinically lame, followed by therapeutic hoof trimming when required. Morbidity was defined as the occurrence of at least one of the following clinical conditions: mastitis, pneumonia, indigestion, or lameness.

Blood Sampling and Isolation and Purification of CD4⁺ T Lymphocytes

Blood was collected once per cow during wk 11 of the treatment period (173 ± 18 DIM). Across an 8-d collection period, 4 cows (2 cows per treatment) were randomly selected to be sampled each day, with each

Table 1. Composition of diets fed to postpartum mid-lactation Holstein cows¹

Item	MPA	MPI
Ingredient, % DM		
Corn silage	28.7 ± 1.2	28.9 ± 1.2
Alfalfa silage	23.3 ± 1.2	23.5 ± 1.1
High-moisture corn	23.3 ± 0.81	23.4 ± 0.88
Cottonseed, fuzzy	4.85 ± 0.49	4.88 ± 0.47
Distillers grains	2.03 ± 0.11	2.04 ± 0.14
Concentrate mix ²	16.8 ± 0.68	17.1 ± 0.81
Additive ³		
Chromium	0.06 ± 0.01	0.06 ± 0.01
RPM	0.07 ± 0.01	0.08 ± 0.01
PBM	0.81 ± 0.03	0.00
RPL	0.00	0.17 ± 0.01
Nutrient, ⁴ DM basis		
NE _L , ⁵ Mcal/d	1.74	1.74
DM, %	53.3 ± 1.07	53.1 ± 1.03
CP, %	17.5 ± 0.06	16.9 ± 0.06
NDF, %	29.5 ± 0.46	29.7 ± 0.47
ADF, %	20.6 ± 0.40	20.7 ± 0.44
Starch, %	27.9 ± 0.68	28.1 ± 0.71
Ether extract, %	3.84 ± 0.03	3.86 ± 0.05
MP, ⁵ g/d		
Targeted	3,111	2,983
Required	2,924	2,840
Observed	2,886	2,669
Balance	−38	−171

¹Mid-lactation Holstein cows were fed either a MP-adequate (MPA) or MP-inadequate (MPI) diet for 12 wk.

²The concentrate was formulated to contain (% as fed): 44.5% soy-bean meal, 19.2% soy hulls, 10.9% expeller meal (SoyPlus, Landus Cooperative), 5.86% calcium carbonate, 4.62% sodium bicarbonate, 2.40% mineral mix (% as fed; 82.12% NaCl; 0.35% Ca; 0.09% S; 0.01% Co; 0.38% monensin; 0.48% Cu; 0.04% I; 0.15% Fe; 1.43% Mn; 0.01% Se; 2.05% Zn; 0.004% biotin; 0.07% diflubenzuron; 1.954 kIU/kg vitamin E; 391 kIU/kg vitamin A; 78 kIU/kg vitamin D₃), 1.46% magnesium oxide (56%), 1.11% urea (46%), 0.57% potassium carbonate, 7.99% tallow (MaxFat, Sanimax Corp.), 0.18% yeast culture (Celmanax Dry, Church & Dwight Co. Inc.), and 0.12% probiotics (Fortress LG, Vita Plus Corp.).

³The additive bags contained chromium (KemTRACE Chromium, Kemin Industries Inc.), rumen-protected Met (RPM; KESSENT Me, Kemin Industries Inc.), porcine blood meal (PBM; Consumers Supply, Sioux City, IA), and rumen-protected Lys (RPL; LysiGEM Extend, Kemin Industries Inc.) in MPA and MPI diets, respectively.

⁴Means of 4 composite samples analyzed for nutrient composition.

⁵Diets were formulated using the NASEM Dairy (NASEM, 2021) model to supply targeted amounts of MP, based on ingredient composition at the start of the study and an assumed DMI of 32 kg/d from previous studies in the same herd. Required and observed MP supplies were subsequently recalculated using actual mean ingredient composition, DMI, and production performance recorded during the 12-wk treatment period.

day serving as a block to account for assay logistics and minimize potential day-to-day variability (Figure 1B). During the single collection, 60 mL was collected from the coccygeal vessel of each cow after the morning milking at approximately 0630 h. Blood was collected in Vacutainer tubes containing K₂EDTA (Vacutainer, Becton Dickinson, Franklin Lakes, NJ). Blood tubes were centrifuged at 1,200 × g for 15 min at room temperature, and the buffy coat was harvested and transferred into a

Table 2. Estimated supply of metabolizable EAA according to diets fed to postpartum mid-lactation Holstein cows¹

Item	MPA	MPI
EAA, ² g/d		
Arg	152	142
His	76	64
Ile	163	156
Leu	267	241
Lys	220	212
Met	71	69
Phe	165	151
Thr	148	139
Trp	38	34
Val	182	164
Lys:Met ³	3.10	3.07

¹Mid-lactation Holstein cows were fed either a MP-adequate (MPA) or MP-inadequate (MPI) diet for 12 wk.

²The estimated supply of EAA, based on NASEM Dairy (NASEM, 2021), was adjusted according to the actual mean DMI recorded throughout the 12-wk treatment period.

³Lys:Met ratio calculated as metabolizable Lys (g)/metabolizable Met (g).

50-mL sterile tube. The buffy coat was diluted with 15 mL of PBS. Peripheral blood mononuclear cells (PBMC) were isolated by density gradient separation using His-topaque-1077 (10771, Sigma-Aldrich, St. Louis, MO) and washed twice in PBS. Red blood cells were removed by hypotonic lysis using double-distilled water. Purified PBMC were resuspended in PBS containing 0.5% BSA (PBS-BSA; Sigma-Aldrich). Peripheral blood mononuclear cells were counted using an automatic cell counter (Countess II FL, Invitrogen, Waltham, MA), and the viability of cells was assessed by diluting cell suspension with 0.08% (vol/vol) solution of trypan blue (T8154, Sigma-Aldrich) in a 1:1 dilution as described previously (Arshad et al., 2024b).

All CD4⁺ T lymphocytes were isolated using magnetic cell separation (Miltenyi Biotec, Bergisch Gladbach, Germany) according to manufacturer's instructions. Briefly, aliquots of 1 × 10⁷ PBMC were incubated for 30 min at room temperature in the dark with 5 µL of anti-CD4 antibody (WS0562B-100, clone ILA11A, Kingfisher Biotech Inc., Saint Paul, MN) in 100 µL of PBS-BSA. Thereafter, PBMC were washed twice with PBS-BSA and incubated for an additional 15 min at 4°C with 20 µL of magnetic-activated cell sorting IgG2a+b magnetic beads (130–047–202, Miltenyi Biotec) diluted in 80 µL of PBS-BSA. Peripheral blood mononuclear cells were washed again with PBS-BSA, and the bead-CD4⁺ T lymphocytes complex was extracted with a LS magnetic-activated cell sorting column (130–042–401, Miltenyi Biotec) according to the manufacturer's instructions. After CD4⁺ T lymphocytes isolation, cells were incubated with Labeling Check Reagent, FITC (130–124–695, Miltenyi Biotec) according to manufacturer's

protocol to check the purity of the cell population using a flow cytometer (Attune NxT Flow Cytometer, Thermo Fisher Scientific, Waltham, MA). The viability of sorted CD4⁺ T lymphocytes was assessed using a 0.08% trypan blue solution in an automatic cell counter (Countess II FL, Invitrogen). The CD4⁺ T lymphocytes were counted to prepare 2 aliquots containing 2.4×10^6 cells in each aliquot for downstream Seahorse analysis.

Evaluating CD4⁺ T Cell Bioenergetics Through T Cell Persistence Assay

The Seahorse analysis was performed to evaluate bioenergetics measures in CD4⁺ T lymphocytes using XFp T Cell Metabolic Profiling Kit (103771–100, Agilent Technologies Inc., Santa Clara, CA) analyzed on a Seahorse XF HS Mini Analyzer. The day before analysis, Agilent Seahorse XFp Extracellular Flux Cartridges (103022–100, Agilent Technologies Inc.) were hydrated according to the manufacturer's instructions and placed in a non-CO₂ incubator overnight at 37°C. Additionally, approximately 5 mL of XF Calibrant medium (103022–100, Agilent Technologies Inc.) for the hydration of each utility plate of cartridge and XFp PDL miniplate (103722–100, Agilent Technologies Inc.) were placed in a non-CO₂ incubator overnight at 37°C. On the day of analysis, the standard Seahorse XF RPMI assay medium was prepared by supplementing 19.4 mL of Seahorse XF RPMI medium (103576–100, Agilent Technologies Inc.) with glucose (103577–100, XF 1.0 M Glucose Solution, Agilent Technologies Inc.), pyruvate (103578–100, XF 100 mM Pyruvate Solution, Agilent Technologies Inc.), and glutamine (103579–100, XF 200 mM Glutamine Solution, Agilent Technologies Inc.) solutions to achieve final concentrations of 10, 1, and 2 mM, respectively, and stored at 37°C throughout the assay as reported previously (Arshad et al., 2024b).

Isolated CD4⁺ T lymphocytes from each MPA- and MPI-fed cow were split into 2 tubes and randomly assigned to incubate with either 250 µL of XF RPMI assay medium to maintain cells in the metabolically NA state or treated with a combination of 140 µL of Seahorse XF RPMI assay medium, 100 µL of phorbol 12-myristate 13-acetate (**PMA**; J63916-M, Thermo Fisher Scientific) at 20 ng/mL, and 10 µL of ionomycin (**IMY**; I24222, Invitrogen) at 1 µg/mL. Cells were incubated for 2 h at 37°C in a bead bath incubator (without controlled CO₂ or humidity conditions), after which the Seahorse assay was initiated immediately. The rationale for incubating CD4⁺ T lymphocytes with a combination of PMA+IMY was to induce immune activation and promote high metabolic activity in CD4⁺ T lymphocytes. This methodology stemmed from the observed increase of cytokine production by immune cells when exposed to this combination

(Ai et al., 2013) and has been reported previously in dairy cows (Arshad et al., 2024b). Each XFp PDL miniplate represented 1 MPA and 1 MPI cow either containing NA or AC state of CD4⁺ T lymphocytes, separately. After a 2 h incubation of CD4⁺ T lymphocytes with assay medium or PMA+IMY at 37°C, 50 µL of CD4⁺ T lymphocyte suspension from each tube was added in triplicate at 4.0×10^5 live cells density/well in the XFp PDL miniplate (Eder et al., 2020; Arshad et al., 2024b). Thereafter, 50 µL of Seahorse XF RPMI assay medium was added to each well, and the moats around each well were filled with 200 µL of warm sterile water. The XFp PDL miniplate was centrifuged at $200 \times g$ for 1 min to allow cells to adhere to the bottom of the wells, and an additional 100 µL of warm XF RPMI assay medium was added in each well. Thereafter, the XFp PDL miniplate was incubated at 37°C in a non-CO₂ incubator for 60 min before performing the T-cell persistence assay. On the day of analysis, the XFp Extracellular Flux Cartridge was removed from the incubator and rehydrated with prewarmed XF Calibrant medium according to manufacturer's instructions. The cartridge was then placed back into the non-CO₂ incubator at 37°C for at least 60 min before adding the port injection substrates. The port injection substrates in the XFp T Cell Metabolic Profiling Kit included complex V inhibitor (oligomycin A), a protonophore uncoupler (BAM 15), and complex I and complex III inhibitors (rotenone and antimycin A). Each reagent was resuspended with 0.5 mL of warm XF RPMI assay medium to achieve final concentrations of 13.5 µM oligomycin A, 25 µM BAM 15, and 5.5 µM rotenone and antimycin A. A total of 25 µL of oligomycin A, BAM 15, and a mixture of rotenone and antimycin A were added to ports A, B, and C, respectively, for each well of the XFp PDL miniplate. The rationale to add these reagents in injection ports was to induce mitochondrial stress in CD4⁺ T lymphocytes and measure the bioenergetic changes in those cells. After injection substrates were added to each port, the cartridge was placed in the XF HS Mini Seahorse (Agilent Technologies Inc.) for calibration. Once the cartridge was calibrated, the XFp PDL miniplate was inserted into the XF HS Mini Seahorse (Agilent Technologies Inc.) machine for analysis to perform a T-cell persistence assay. The assay was conducted in triplicate, and 2 wells within each PDL miniplate were designated for quality-control (lacking CD4⁺ T lymphocytes) purposes. The rationale for including quality-control wells was to monitor baseline oxygen and pH drift, as well as to verify sensor calibration and stability during each run. The average signal from these wells was used to correct the background of the CD4⁺ T cell-containing wells at corresponding time points. The overall intra- and inter-assay CV across all phases of the T-cell persistence assay for OCR, ECAR, and PER were below 10%.

Data were exported and quality of data examined using the raw Excel (Microsoft Corp.) outputs and Seahorse Wave Pro Software (version 10.1.0, Agilent Technologies Inc., 2021). During this assay, mitochondrial function kinetics were recorded in real time, measuring oxygen consumption rate (OCR; pmol/min) as a marker for oxidative phosphorylation, whereas cytoplasmic pH changes were monitored in real time to monitor extracellular acidification rate (ECAR; pmol/min) as an indicator of cellular glycolysis. Additionally, proton efflux rate (PER; pmolH⁺/min) was also determined to represent a quantitative assessment of extracellular acidification. The measures of bioenergetics were calculated using Seahorse Analytics (version 1.0.0–570, Agilent Technologies Inc., 2023) and included basal respiration (pmol/min), maximal respiration (pmol/min), SRC (pmol/min), mitochondrial ATP production rate (pmol/min), glycolytic ATP production rate (pmol/min), and total ATP production rate (pmol/min). The bioenergetic measures for each plate were averaged across each triplicate.

Statistical Analyses

The experiment followed a split-plot design. The whole plot represented 2 types of diet (MPA vs. MPI), whereas the sub-plot was based on activation status during cell analysis (NA vs. AC; 16 blocks). Each block represented 2 Seahorse mini plates (i.e., 1 plate for NA and 1 plate for metabolically AC CD4⁺ T lymphocytes), and each plate contained 1 cow from each treatment. Normality of residuals and homogeneity of variance were examined for each continuous dependent variable after fitting the statistical model. Some responses (OCR, ECAR, and PER) violated the assumptions of normality and were subjected to power transformation according to the Box–Cox procedure (Box and Cox, 1964) using PROC TRANSREG in SAS 9.4 (SAS Institute Inc., Cary, NC); however, the interpretation of data did not change after transformation. Therefore, analysis of untransformed data was used, and the LSM and respective SEM were presented on the original scale.

Data were analyzed by mixed-effects models using the MIXED procedure of SAS 9.4 (SAS Institute Inc.). Data pertaining to single time point variables were analyzed using the following model:

$$Y_{ijk} = \mu + D_i + AS_j + D_i \times AS_j + B_k + D_i \times B_k + \varepsilon_{ijk},$$

where Y_{ijk} is the dependent variable, μ is the overall mean, D_i is the fixed effect of diet (MPA vs. MPI), AS_j is the fixed effect of activation status (NA vs. AC), $D_i \times AS_j$ is the interaction between diet and activation status, B_k is the random effect of block, $D_i \times B_k$ is the random effect of interaction between diet and block, and ε_{ijk} is the

random residual. Data pertaining to repeated measures were analyzed using the following model:

$$Y_{ijklm} = \mu + D_i + AS_j + D_i \times AS_j + T_k + D_i \times T_k + AS_j \times T_k + D_i \times AS_j \times T_k + B_l + B_l(D_i) + AS_j \times B_l(D_i) + C_m(D_i \times AS_j \times B_l) + \varepsilon_{ijklm},$$

where Y_{ijklm} is the dependent variable, μ is the overall mean, D_i is the fixed effect of diet (MPA vs. MPI), AS_j is the fixed effect of activation status (NA vs. AC), $D_i \times AS_j$ is the interaction between diet and activation status, T_k is the fixed effect of time within the Seahorse assay, $D_i \times T_k$ is the interaction between diet and time, $AS_j \times T_k$ is the interaction between activation status and time, $D_i \times AS_j \times T_k$ is the 3-way interaction between diet, activation status, and time, whereas the statistical models included 3 random statements based on previously described recommendations (St-Pierre, 2007): (1) B_l is the random effect of block, (2) $B_l(D_i)$ is the random effect of block nested within diet, and (3) $AS_j \times B_l(D_i)$ is the random effect of interaction between activation status and block nested within diet. Additionally, in all statistical models with repeated measures, $C_m(D_i \times AS_j \times B_l)$ represents the cow nested within an interaction between diet, activation status, and block as a proper error term for assessing the effects of diet and activation status as described previously (St-Pierre, 2007), and ε_{ijklm} is the random effect of residual. Although the response variables had equal spacing between measurements, multiple covariance structures were tested. Based on model convergence and fit statistics, the autoregressive 1 structure provided the best description of the data and was therefore used for the repeated-measures analyses. When an interaction between diet and time resulted in $P \leq 0.05$, means of both diets at different time points were separated using the SLICE command of SAS 9.4 (SAS Institute Inc.).

Binary data, including health events, were analyzed by generalized linear mixed-effects models using logistic regression with the GLIMMIX procedure of SAS, fitting a binary distribution. The statistical models included the fixed effects of treatment. For analyses of incidence of pneumonia and indigestion, Fisher's exact test was used because of the low frequency of events.

In all mixed-effects models, the Kenward–Roger method was used to approximate the denominator df for the F -tests. Data are reported as LSM and SEM. Evidence of statistical significance against the null hypothesis was considered at $P \leq 0.05$, and tendencies were considered at $0.05 < P \leq 0.10$.

RESULTS

Data on productive performance, ECM:feed, and nutrient partitioning are presented in the companion article (Arshad et al., 2026). Based on observed DMI, the actual MP supply was 2,886 and 2,669 g/d for the MPA and MPI treatments, respectively (Arshad et al., 2026). All 32 cows remained healthy and contributed data, with no recorded cases of mastitis, pneumonia, lameness, or indigestion for at least 40 d before evaluating metabolic function in CD4⁺ T lymphocytes.

Viability and Purity of PBMC and CD4⁺ T Lymphocytes

The mean $\pm$ SD viability of PBMC and CD4⁺ T lymphocytes was 92.0% $\pm$ 5.3% and 92.1% $\pm$ 3.8%, respectively, in MPA- and MPI-fed cows. The purity of CD4⁺ T lymphocytes after isolation from PBMC was 91.5% $\pm$ 0.8% and 91.7% $\pm$ 0.8% in MPA- and MPI-fed cows, respectively, and dot plots from representative MPA- and MPI-fed cows are presented in Figure 2A–D.

Effects of MP Supply on OCR of CD4⁺ T Lymphocytes

Cows fed the MPI diet exhibited lower ($P = 0.01$) overall OCR than those fed the MPA diet (Figure 3A; Table 3). Meanwhile, overall OCR was reduced ($P = 0.05$) in metabolically AC CD4⁺ T lymphocytes compared with NA cells (Figure 3A; Table 3). No interaction between diet and activation status was detected (Figure 3A; Table 3), whereas interactions between diet and time ($P < 0.0001$) and between activation status and time ($P < 0.0001$) were observed, reflecting dynamic changes in overall OCR over time within the assay (Figure 3A; Table 3). Following oligomycin A injection, basal respiration decreased ($P < 0.01$) and reached a nadir across all treatments (Figure 3A). The addition of BAM 15 induced maximal respiration, with MPA-NA-treated CD4⁺ T lymphocytes achieving the highest ($P < 0.01$) OCR and MPI-AC cells the lowest ($P < 0.01$; Figure 3A). After rotenone and antimycin A injection, OCR returned to minimal levels, confirming effective mitochondrial respiration inhibition (Figure 3A).

Effects of MP Supply on ECAR or PER of CD4⁺ T Lymphocytes

In metabolically AC CD4⁺ T lymphocytes, overall ECAR and PER were greater ($P < 0.0001$) compared with NA cells (Figure 3B and 3C; Table 3). No interactions between diet and activation status were detected for overall ECAR or PER across treatments (Figure 3B and

3C; Table 3). However, interactions ($P \leq 0.01$) between diet and time and between activation status and time were observed for overall ECAR and PER, reflecting dynamic changes in glycolytic activity and proton efflux over time within the assay (Figure 3B and 3C; Table 3). At baseline, ECAR and PER values were low but not significantly different across both treatments (Figure 3B and 3C). Following oligomycin A injection, both ECAR and PER increased ($P < 0.001$) steadily, with AC lymphocytes showing a more pronounced response than NA cells (Figure 3B and 3C). The addition of BAM 15 further elevated ($P < 0.001$) ECAR and PER, particularly in MPA-fed cows, whereas MPI-fed cows exhibited a progressively attenuated response (Figure 3B and 3C). After rotenone and antimycin A treatment, ECAR and PER declined, confirming suppression of mitochondrial and glycolytic activity (Figure 3B and 3C).

Effects of MP Supply on Mitochondrial Respiration of CD4⁺ T Lymphocytes

Basal respiration was not affected by either diet or activation status (Figure 4A). Feeding the MPI diet decreased ($P = 0.006$) maximal respiration in both metabolically NA (MPA: 55.4 vs. MPI: 41.9 $\pm$ 5.8 pmol/min) and AC (MPA: 42.4 vs. MPI: 29.0 $\pm$ 5.8 pmol/min) CD4⁺ T lymphocytes (Figure 4B). Moreover, maximal respiration in metabolically NA CD4⁺ T lymphocytes was greater ($P = 0.0008$) than in AC cells (48.7 vs. 35.7 $\pm$ 5.8 pmol/min; Figure 4B). Feeding the MPI diet also reduced ($P = 0.002$) SRC in both metabolically NA (MPA: 37.2 vs. MPI: 26.2 $\pm$ 4.6 pmol/min) and AC (MPA: 25.9 vs. MPI: 12.9 $\pm$ 4.6 pmol/min) CD4⁺ T lymphocytes (Figure 4C). Additionally, SRC in metabolically NA CD4⁺ T lymphocytes was greater ($P = 0.0004$) compared with AC counterparts (31.7 vs. 19.4 $\pm$ 4.6 pmol/min; Figure 4C).

Effects of MP Supply on ATP Production Rates in CD4⁺ T Lymphocytes

Basal mitochondrial, glycolytic, and total ATP production rates were not affected by diet (Figure 5A to 5C). However, glycolytic ATP production rate was higher ($P < 0.001$) in the metabolically AC CD4⁺ T lymphocytes compared with NA cells (Figure 5B). For cows fed the MPA diet, glycolytic ATP production rate increased ($P < 0.001$) from 30.9 $\pm$ 3.96 pmol/min in NA cells to 52.2 $\pm$ 3.96 pmol/min in AC cells (Figure 5B). Similarly, for cows fed the MPI diet, glycolytic ATP production rate increased ($P < 0.001$) from 31.4 $\pm$ 3.96 pmol/min in NA cells to 54.7 $\pm$ 3.96 pmol/min in AC cells (Figure 5B). Total ATP production rate was also higher ($P = 0.01$) in AC compared with NA CD4⁺ T lymphocytes (Figure 5C).

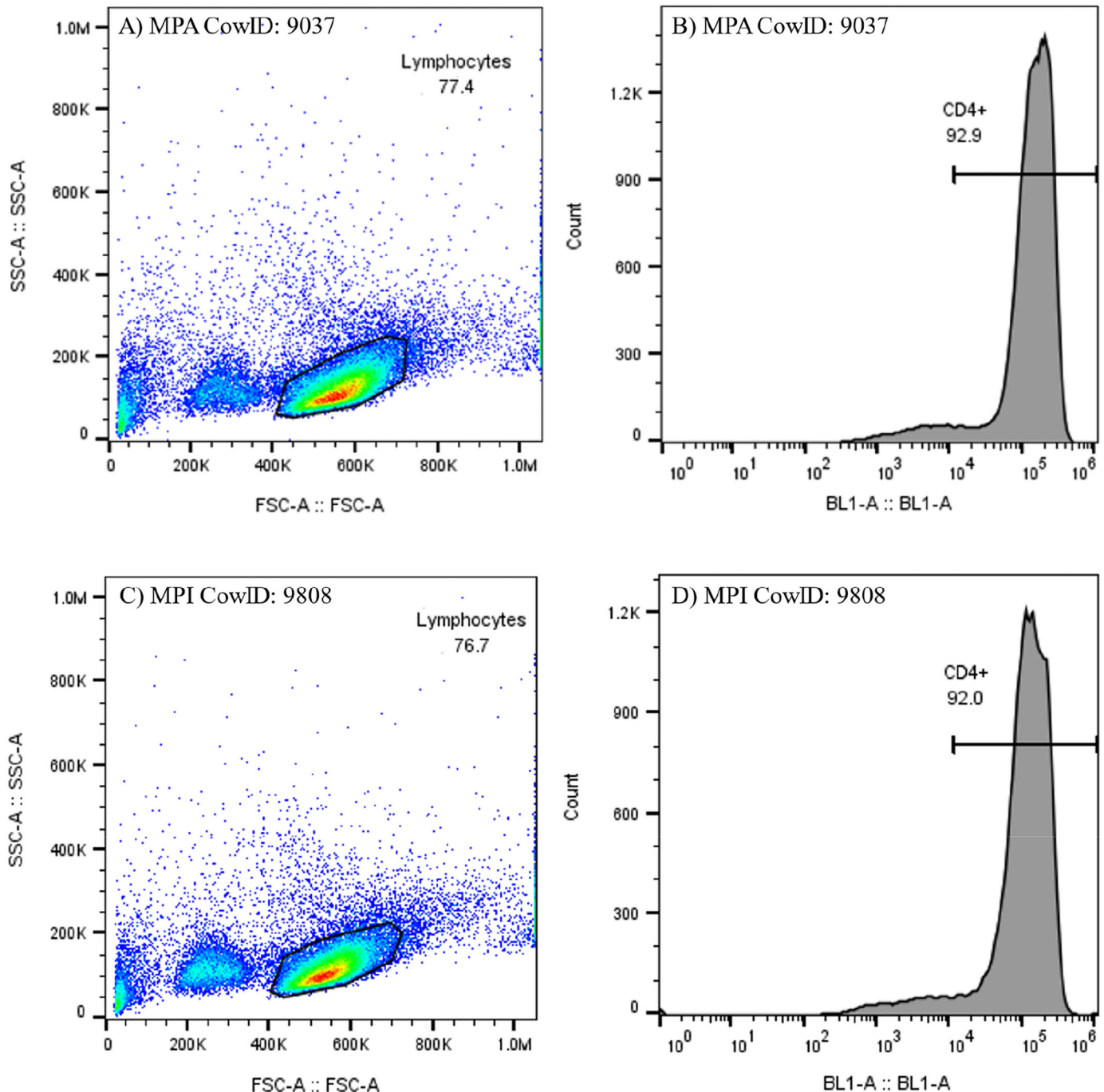


Figure 2. Purity of CD4⁺ T lymphocytes after magnetic isolation from representative cows fed either a MP-adequate (MPA; panels A and B) or MP-inadequate (MPI; panels C and D) diet. All CD4⁺ T cells were isolated from peripheral blood mononuclear cells using magnetic cell separation and analyzed by flow cytometry. Lymphocytes were first gated based on forward scatter (FSC) and side scatter (SSC) parameters (panels A and C), and thereafter, CD4⁺ T lymphocytes were identified within the lymphocyte gate using FITC-conjugated anti-CD4 antibody and detected in channel BL1-A (panels B and D). The final purity of CD4⁺ T cells was 93% and 92% in the MPA- or MPI-fed cows, respectively.

Effects of MP Supply on Health Events

No differences were observed in health outcomes between treatments, as reducing the supply of MP in cows

fed the MPI diet did not affect the incidence of mastitis, lameness, indigestion, pneumonia, or overall morbidity throughout the 12 wk of treatment period (Table 4). The overall incidence of mastitis was 9.4%, with 8.9% and

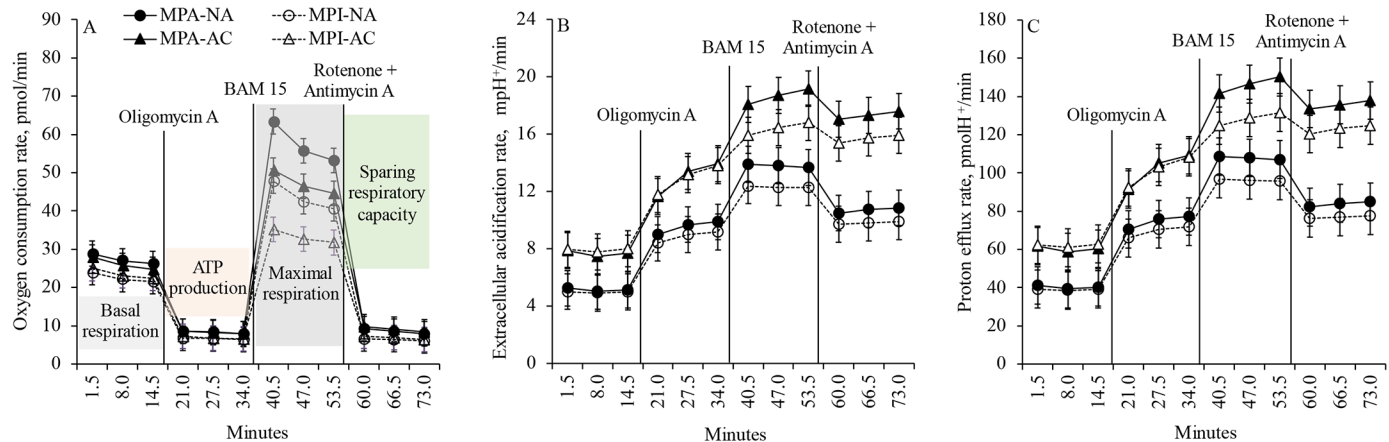


Figure 3. Effects of feeding MP-adequate (MPA) or MP-inadequate (MPI) diets on metabolic function of CD4⁺ T lymphocytes, assessed according to activation status (AS = nonactivated [NA] or activated [AC]) in mid-lactation dairy cows. Oxygen consumption rate (OCR; panel A), extracellular acidification rate (ECAR; panel B), and proton efflux rate (PER; panel C) are presented. Mitochondrial and glycolytic functional kinetics in CD4⁺ T lymphocytes were recorded in real-time measuring OCR and ECAR, respectively, under basal conditions and in response to complex V inhibitor (oligomycin A), a protonophore uncoupler (BAM 15), and complex I and complex III inhibitors (rotenone and antimycin, respectively A). Panel A: effects of diet ($P = 0.01$), AS ($P = 0.05$), interaction between diet and AS ($P = 0.82$), interaction between diet and time within the assay ($P < 0.0001$), and interaction between AS and time within the assay ($P < 0.0001$). Panel B: effects of diet ($P = 0.30$), AS ($P < 0.0001$), interaction between diet and AS ($P = 0.91$), interaction between diet and time within the assay ($P = 0.01$), and interaction between AS and time within the assay ($P < 0.0001$). Panel C: effects of diet ($P = 0.30$), AS ($P < 0.0001$), interaction between diet and AS ($P = 0.90$), interaction between diet and time within the assay ($P = 0.01$), and interaction between AS and time within the assay ($P < 0.0001$). Error bars depict SEM.

6.3% for MPA and MPI-fed cows, respectively. Pneumonia incidence was 6.3% in both dietary groups. Indigestion occurred in 3.13% of cows overall, with 6.3% in MPA-fed cows and no cases in MPI-fed cows. Lameness incidence was 6.3% for both groups. Overall morbidity did not differ between treatments (MPA = 21.7% vs. MPI = 18.8% $\pm$ 8.4%).

DISCUSSION

Amino acids are essential regulators of immune cell fate and function, particularly in active lymphocytes undergoing rapid proliferation (Palsson-McDermott and O'Neill, 2013). Upon activation, lymphocytes markedly increase their demand for glucose and AA to support protein synthesis, nucleotide production, and energy metabolism (Palsson-McDermott and O'Neill, 2013;

Castellano and Molinier-Frenkel, 2020). This metabolic surge is tightly linked to immune effector functions and cell survival. Key pathways such as glycolysis, oxidative phosphorylation, and cytokine production depend on adequate AA availability (Castellano and Molinier-Frenkel, 2020). Depletion of specific AA can impair lymphocyte proliferation, redox homeostasis, and cytokine synthesis, thereby weakening immune responses even in the absence of overt disease (Calder, 2013). In particular, dietary protein deficiency reduces the circulating availability of critical AA including Arg, Cys, Gln, His, Lys, Met, and Trp, thereby compromising immune competence (Wu et al., 1999; Kim et al., 2007). Supporting the observed metabolic effects, the diet formulated with a suboptimal supply of MP in this study limited the availability of several EAA, including His, Leu, Trp, Val, and Arg. Restricting multiple indispensable AA

Table 3. Effects of feeding diets differing in supply of MP on metabolic function of CD4⁺ T lymphocytes in mid-lactation dairy cows¹

Item, ³ pmol/min	MPA		MPI		SEM	P -value ²				
	NA	AC	NA	AC		D	AS	D $\times$ AS	D $\times$ T	AS $\times$ T
OCR	25.4	22.7	19.8	17.6	2.3	0.01	0.05	0.82	<0.0001	<0.0001
ECAR	9.8	14.2	9.0	13.2	1.1	0.30	<0.0001	0.91	0.01	<0.0001
PER	76.5	111	70.4	104	9.0	0.30	<0.0001	0.90	0.01	<0.0001

¹Mid-lactation Holstein cows were fed either a MP-adequate (MPA) or MP-inadequate (MPI) diet for 12 wk to determine the metabolic function of CD4⁺ T lymphocytes, assessed according to activation status (AS = nonactivated [NA] or activated [AC]).

²Effects of diet (D; MPA vs. MPI), AS (NA vs. AC), interaction between D and AS, interaction between D and time (T) within the Seahorse assay, and interaction between AS and T within the Seahorse assay.

³OCR = oxygen consumption rate; ECAR = extracellular acidification rate; PER = proton efflux rate.

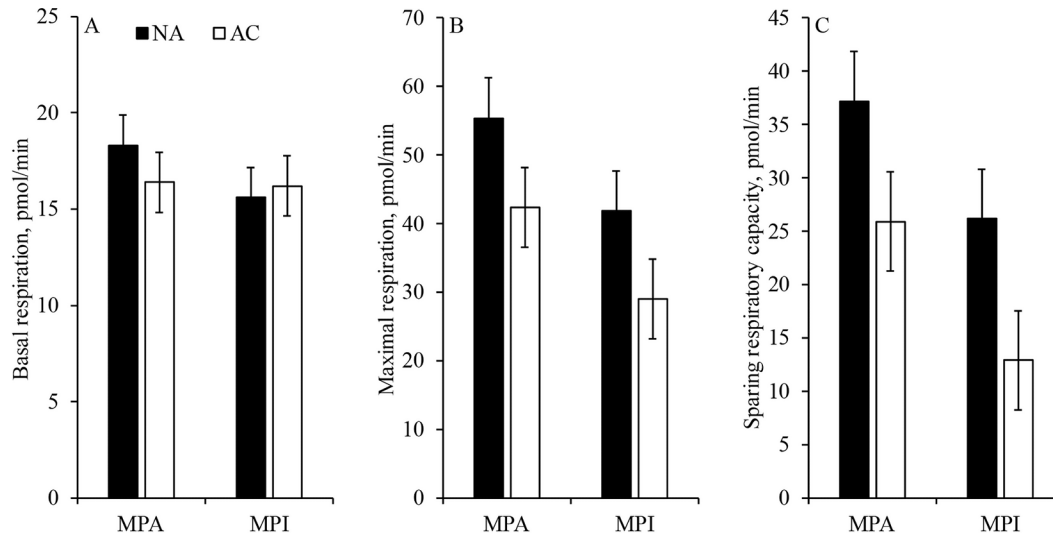


Figure 4. Effects of feeding MP-adequate (MPA) or MP-inadequate (MPI) diets on mitochondrial respiration of CD4⁺ T lymphocytes, assessed according to activation status (AS = nonactivated [NA] or activated [AC]) in mid-lactation dairy cows. Basal respiration (panel A), maximal respiration (panel B), and sparing respiratory capacity (panel C) are presented. Panel A: effects of diet ($P = 0.36$), AS ($P = 0.50$), and the interaction between diet and AS ($P = 0.24$). Panel B: effects of diet ($P = 0.006$), AS ($P = 0.0008$), and the interaction between diet and AS ($P = 0.99$). Panel C: effects of diet ($P = 0.002$), AS ($P = 0.0004$), and the interaction between diet and AS ($P = 0.73$). Error bars depict SEM.

simultaneously imposes a broader metabolic bottleneck on immune cells, potentially weakening mitochondrial adaptability, limiting substrate availability for biosynthesis, and ultimately reducing the energetic flexibility of CD4⁺ T lymphocytes. It is noteworthy that the reduction in MP supply in MPI-fed cows (~171 g/d) represents a moderate deficit and is substantially smaller than the 500 to 700 g/d shortfalls typically observed in early lactation; therefore, the immunometabolic effects reported in this study should be interpreted within the context of a controlled nutrient insufficiency rather than a severe transition-period challenge.

To assess how AA availability affects T-cell metabolic fitness, we measured OCR, ECAR, and PER in real time using extracellular flux analysis, providing a quantitative framework to evaluate energetic capacity and flexibility

under basal and stress conditions. Oxygen consumption rate reflects mitochondrial respiration and the ability of T cells to generate ATP via oxidative phosphorylation, whereas ECAR serves as a proxy for glycolytic flux, which is critical during rapid proliferation and effector activation (Eder et al., 2020). Proton efflux rate complements these measures by integrating glycolytic and mitochondrial proton fluxes, offering a more comprehensive view of metabolic activity. In CD4⁺ T cells, activation through T-cell receptor signaling or indirect stimulation via antigen-presenting cells increases energetic demand, ATP turnover, and dynamic shifts in both glycolysis and mitochondrial respiration. By quantifying OCR, ECAR, and PER under NA and AC conditions, we evaluated T-cell metabolic flexibility, which reflects the ability of those cells to meet sudden energy demands, support

Table 4. Effects of feeding diets differing in supply of MP on the incidence of diseases in mid-lactation Holstein cows

Item	Incidence (n/n)	Treatment ¹		SEM	P-value ²	
		MPA	MPI		Diet	
Mastitis, %	9.38 (3/32)	8.90	6.26	5.23	0.55	
Pneumonia, ³ %	6.25 (2/32)	6.25	6.25	—	0.99	
Indigestion, ³ %	3.13 (1/32)	6.25	0	—	0.96	
Lameness, %	6.25 (2/32)	6.25	6.25	4.28	0.99	
Morbidity, %	21.9 (7/32)	21.7	18.8	8.35	0.67	

¹Mid-lactation Holstein cows were fed either a MP-adequate (MPA) or MP-inadequate (MPI) diet for 12 wk.

²Effect of diet (MPA vs. MPI).

³Data were analyzed by Fisher's exact test because of the low frequency of events to determine the effect of treatment.

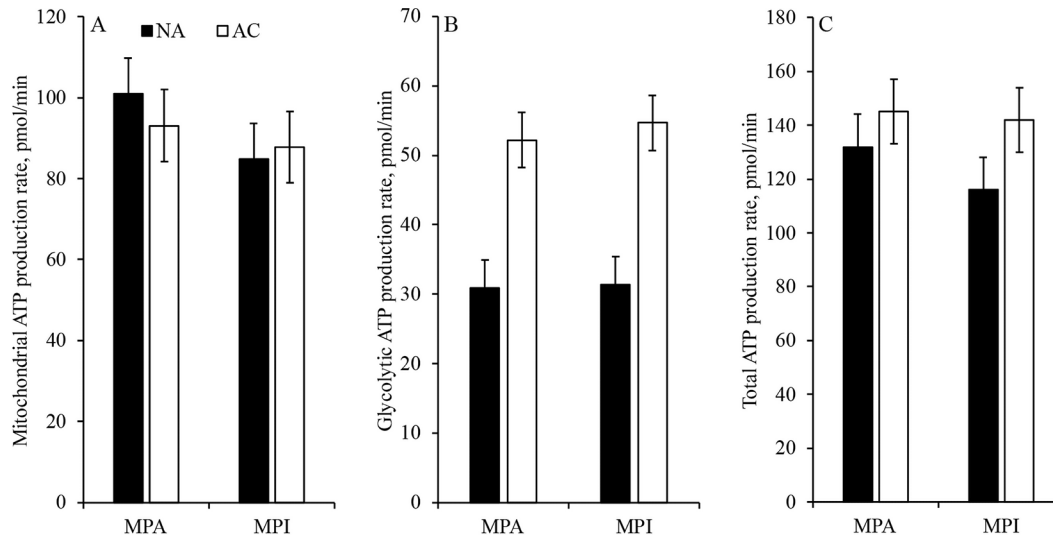


Figure 5. Effects of feeding MP-adequate (MPA) or MP-inadequate (MPI) diets on ATP production rates of CD4⁺ T lymphocytes, assessed according to activation status (AS = nonactivated [NA] or activated [AC]) in mid-lactation dairy cows. Mitochondrial (panel A), glycolytic (panel B), and total ATP production rates (panel C) are presented. Panel A: effects of diet ($P = 0.21$), AS ($P = 0.60$), and the interaction between diet and AS ($P = 0.29$). Panel B: effects of diet ($P = 0.66$), AS ($P < 0.001$), and the interaction between diet and AS ($P = 0.74$). Panel C: effects of diet ($P = 0.41$), AS ($P = 0.01$), and the interaction between diet and AS ($P = 0.39$). Error bars depict SEM.

biosynthesis, and maintain redox balance. This capacity is essential for effective adaptive immune responses, including proliferation, cytokine production, and differentiation into specialized subsets such as Th1, Th17, and Treg cells (Pearce et al., 2013; O'Neill et al., 2016).

In this study, we investigated the effects of MP adequacy or deficiency on metabolic function under mitochondrial stress in NA and AC CD4⁺ T lymphocytes from high-producing mid-lactation Holstein dairy cows. Cows receiving the MPI diet exhibited markedly reduced OCR in CD4⁺ T lymphocytes compared with MPA-fed cows. This is consistent with the known role of AA in supporting mitochondrial and glycolytic metabolism, rather than being limited to protein synthesis (Kelly and Pearce, 2020). Amino acids such as Arg, Glu, Cys, and Met are critical for T-cell metabolic function because they provide substrates and cofactors that directly support specific aspects of energy metabolism and redox balance. For example, Glu and Arg can feed into the tricarboxylic acid cycle to sustain mitochondrial respiration and maximal ATP production (Yoo et al., 2020), whereas Met and Cys contribute to the synthesis of glutathione, buffering reactive oxygen species and protecting mitochondrial integrity (Garg et al., 2011). These AA collectively influence glycolytic flux and oxidative phosphorylation, thereby supporting both basal and maximal respiratory capacity as well as the persistence of CD4⁺ T lymphocytes under activated conditions (Wu et al., 1999; Kim et al., 2007; Wang and Green, 2012). The dataset observed in this study indicates that CD4⁺ T cells from MPA cows had higher maximal respiration and SRC,

regardless of activation status, as described in schematic Figure 6. Spare respiratory capacity reflects mitochondrial adaptability and energetic reserve, defined as the ability of mitochondria to increase ATP production via oxidative phosphorylation in response to acute energy or biosynthetic demands (Sena et al., 2013; Buck et al., 2016). This capacity depends on both the functional integrity of the electron transport chain and the availability of electron donors such as NADH and FADH₂. When T cells face sustained or high-energy demands, mitochondria with reduced SRC cannot sufficiently elevate ATP output, causing electrons to accumulate and leak to molecular oxygen, generating reactive oxygen species (Sena et al., 2013; Buck et al., 2016). Elevated reactive oxygen species can damage mitochondrial proteins, lipids, and DNA, further compromising respiratory function and T cell performance, even if cells appear phenotypically normal under basal conditions (Peng et al., 2021). Amino acids serve not only as metabolic substrates but also as key regulators of immune cell metabolism. For instance, Leu activates the mTORC1 signaling pathway, thereby enhancing protein synthesis, glycolytic flux, and T cell proliferation (O'Neill et al., 2016). Similarly, Glu and Ser support nucleotide synthesis, antioxidant production, and chromatin remodeling through one-carbon metabolism and acetyl-CoA generation (O'Neill et al., 2016; Ron-Harel et al., 2019). These interconnected pathways are known to regulate the metabolic programming of naïve T cells, which in turn influences their potential to differentiate into proinflammatory (e.g., Th1, Th17) or regulatory (Treg) subsets, as demonstrated in human and

murine models (Guglani and Khader, 2010; Geiger et al., 2016). Consistent with this, depletion of AA such as Glu has been shown *in vitro* to impair Th17 cell differentiation and reduce inflammatory cytokine production (Guglani and Khader, 2010; Geiger et al., 2016). Although we did not characterize specific CD4⁺ T-cell subsets in this study, and therefore cannot exclude potential differences in subset composition or baseline activation status before the Seahorse assay, our results likely reflect the overall metabolic capacity of circulating CD4⁺ T cells in response to differences in MP supply. It is important to note that subset of T cells identification would require cytokine profiling or intracellular staining (e.g., IFN γ , IL-17, FoxP3), which was beyond the scope of the current study. Nevertheless, healthy mid-lactation cows are expected to have circulating CD4⁺ T cells predominantly composed of naïve or central-memory cells with limited proliferation before antigen exposure (Sterle et al., 2024), whereas early lactation is characterized by increased proliferation and differentiation of CD4⁺ T cells (Fernandez-Wallace et al., 2026). Given this physiological context, large baseline differences in subset abundance are unlikely in the present study, and we therefore interpret the observed differences in mitochondrial fitness primarily as a consequence of MP supply rather than pre-existing polarization.

Following activation, lymphocytes reprogram metabolism to favor aerobic glycolysis, a phenomenon known as the Warburg effect, which generates rapid ATP, NADPH, and biosynthetic intermediates required for proliferation and effector functions (Warburg, 1956; Arshad et al., 2024b). In our study, CD4⁺ T cells in NA or AC states from MPI-fed cows exhibited an attenuated metabolic switch under mitochondrial stress, evidenced by reductions of ~22% in OCR, ~27% in maximal respiration, and ~40% in SRC relative to MPA-fed cows. Glycolytic activity was also reduced in MPI-fed cows, with ECAR and PER decreased by ~8% to 9% during the T-cell persistence assay. These changes are biologically meaningful and comparable to other T-cell models: Human CD4⁺ T cells typically show 15% to 40% shifts in OCR or ECAR when comparing activated versus resting states (Jones et al., 2019), whereas bovine studies report activation-induced changes of 15% to 33% in maximal respiration and 25% to 40% in SRC across lactation stages (Eder et al., 2020; Fernandez-Wallace et al., 2026). Collectively, these data indicate that suboptimal MP supply can impair metabolic flexibility in CD4⁺ T lymphocytes, even when initial glycolytic activation remains largely intact. This impairment may be due, in part, to insufficient levels of AA such as Leu, which were lower in the MPI diet because PBM, a rich source of Leu, was omitted from this treatment. Leucine and Ile have been shown to potentially enhance the translocation of

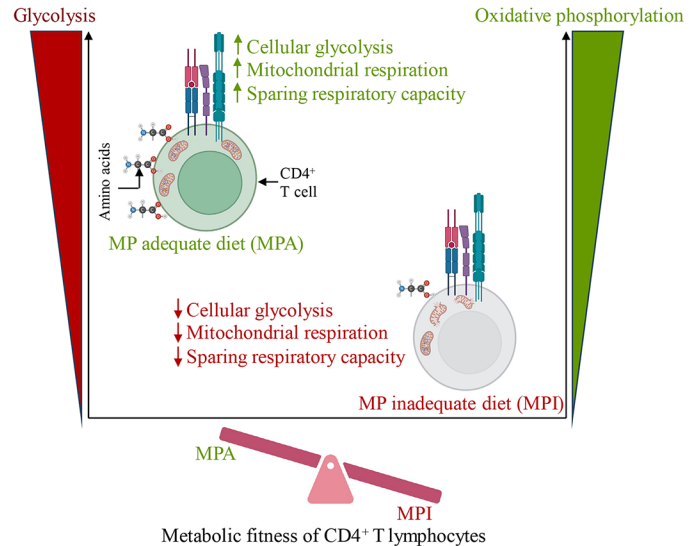


Figure 6. Schematic representation of the effects of MP supply on metabolic function of CD4⁺ T lymphocytes in dairy cows. Holstein cows were fed either a MP-adequate diet (MPA) or MP-inadequate (MPI) diet for 12 wk. The CD4⁺ T lymphocytes from MPI-fed cows exhibited reduced mitochondrial respiration including maximal respiration and sparing respiratory capacity, and decreased cellular aerobic glycolysis compared with MPA. These results may partly explain how reduced metabolic fitness of CD4⁺ T cells could influence immune function during periods of nutritional imbalance in dairy cows.

glucose transporters (e.g., GLUT1) to the cell membrane and provide CoA intermediates for the tricarboxylic acid cycle (Yaqoob and Calder, 1997; Nishitani et al., 2005), which could contribute to supporting energy metabolism in MPA-fed cows, although we did not directly quantify these transporters herein. Reduced spare SRC in CD4⁺ T cells from MPI-fed cows indicates a diminished ability to dynamically adjust mitochondrial and glycolytic activity in response to acute energy demands. In the broader context, it is plausible to speculate that a suboptimal supply of AA, by compromising immune cell function, could ultimately impair the animal's ability to mount effective immune responses, increasing susceptibility to infections, prolonging recovery, and potentially affect overall productivity and health in dairy systems.

Despite reduced SRC in MPI-fed cows, mitochondrial, glycolytic, and total ATP production rates remained unaffected across diets. This apparent disconnect may be explained by several hypothetical mechanisms that were not directly assessed in this study; however, they offered as potential explanations for the reduced SRC despite maintained ATP production. First, steady-state ATP demand in T cells could be met through baseline activity of electron transport chain and substrate-level phosphorylation, which may persist even with limited maximal respiration (Spinelli and Haigis, 2018). Second, nutrient stress under MPI could potentially activate metabolic checkpoints such as AMP-activated protein kinase (AMPK).

Although we did not directly measure AMPK activity in our CD4⁺ T cells, AMPK has been shown in other species to enhance mitochondrial efficiency and reduce anabolic ATP consumption, thereby helping to preserve cellular energy balance (Blagih et al., 2012). Third, although AA are crucial for mitochondrial protein synthesis and anaplerosis, ATP production in T cells depends on the activation state: quiescent T cells primarily oxidize fatty acids, whereas activated T cells rely on glucose-derived substrates to support oxidative phosphorylation (Pearce et al., 2013). A limitation of the present study is that glucose was not measured in the assay medium, which would have provided additional insights into substrate utilization; this should be addressed in future analyses. Last, although activated T lymphocytes predominantly generate ATP via aerobic glycolysis (Warburg effect), mitochondria contribute to biosynthesis and provide a reserve capacity that may be critical under conditions of high energetic demand or nutrient stress. Thus, declines in mitochondrial reserve do not necessarily translate into immediate deficits in ATP production, but may indicate vulnerability under stress (Chang et al., 2013; Buck et al., 2015).

Given the critical role of T cells in coordinating adaptive immune responses, compromised mitochondrial function may have downstream effects on overall immune competence. The findings observed in this study gain further relevance in the context of periparturient dairy cows, where immune dysregulation is common (Bradford and Contreras, 2024). The transition period is characterized by negative energy balance, systemic inflammation, and immunosuppression (Drackley, 1999; Bradford and Contreras, 2024). During this time, the demand for protein and energy increases sharply due to rising milk production, whereas DMI increases more gradually, resulting in a mismatch that contributes to negative energy balance (Drackley, 1999). Suboptimal MP supply in such scenarios not only reduces production performance (Larsen et al., 2014; Zang et al., 2021; Westhoff et al., 2024), but may also exacerbate immune dysfunction, contributing to the onset of clinical disease. A recent study evaluated whether increased MP supply during the transition period altered the inflammatory response to intravenous LPS challenge but reported no effect of MP supply on plasma cytokines (TNF, IL-10), serum haptoglobin, or whole blood leukocyte counts following LPS challenge (Westhoff et al., 2025a). It was speculated that AA diverted toward milk synthesis, coupled with energy limitations, may have obscured any immune-modulatory effects of MP supply (Westhoff et al., 2025a). In contrast, our findings indicate that suboptimal supply of AA can compromise T cell bioenergetics and mitochondrial adaptability, at least under mid-lactation conditions. This divergence may reflect differences in physiological

stage, immune cell types assessed, or challenge model sensitivity. Together, these results emphasize the importance of metabolic context and methodological approach when evaluating nutrient–immune interactions. The immunometabolic axis is sensitive to nutrient availability, and disruptions in mitochondrial energetics may prime immune cells for failure under even moderate infectious challenges (O'Neill et al., 2016; Hotamisligil, 2017; Ron-Harel et al., 2019).

Nevertheless, no differences were observed in clinical health outcomes between MPA- and MPI-fed cows throughout the 12-wk treatment period (Arshad et al., 2026), which is not unexpected given that mid-lactation cows are generally healthy and at low risk of clinical disease. Importantly, the study was designed and powered to assess cellular metabolism rather than disease incidence; therefore, metabolic changes may precede overt clinical signs. Indeed, reductions in mitochondrial respiration under induced mitochondrial stress, along with a reduced SRC in immune cells, may serve as early indicators of physiological vulnerability. Previous studies in other species have shown that low SRC in immune cells correlates with impaired vaccine responses, increased susceptibility to infection, and chronic inflammation (Hotamisligil, 2017). Translating these findings to dairy cows suggests that compromised T cell energetics, driven by protein insufficiency, could represent a latent risk factor for immunological failure under stress or pathogen exposure. Future research should include *in vivo* immune challenges, such as LPS or pathogen exposure, to determine whether impaired cellular metabolism translates into altered immune responses in mid-lactation dairy cows. Larger, longitudinal trials with clinical endpoints are also needed to establish the threshold of MP deficiency that affects both cellular function and animal health. Exploring the potential of targeted AA supplementation to support T cell energetics and improve immune function could inform precision-nutrition strategies for critical periods such as early lactation or disease challenge.

CONCLUSIONS

Feeding suboptimal supply of MP compromised the metabolic function and bioenergetic capacity of CD4⁺ T lymphocytes in mid-lactation dairy cows, mainly reducing maximal respiration, SRC, and aerobic glycolysis over time within the assay. Despite these cellular impairments, mitochondrial, glycolytic, or total ATP production rates and morbidity were unaffected in cows fed an MPI diet. Overall, the findings highlight that while an MPI diet reduces mitochondrial adaptability and impairs metabolic switching, basal mitochondrial or glycolytic ATP production remains buffered by intrinsic T cell flexibility and compensatory metabolic strategies. Future

research should explore MP requirements under conditions of immune activation induced by controlled in vivo challenges, and conduct larger-scale studies powered to evaluate clinical outcomes.

NOTES

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Nonstandard abbreviations used: AC = activated; AS = activation status; ECAR = extracellular acidification rate; IMY = ionomycin; MPA = MP adequate; MPI = MP inadequate; NA = nonactivated; OCR = oxygen consumption rate; PBM = porcine blood meal; PBMC = peripheral blood mononuclear cells; PBS-BSA = PBS mixed with BSA; PER = proton efflux rate; PMA = phorbol 12-myristate 13-acetate; RP = rumen-protected; RPL33, RPL66, RPL100 = diets where 33%, 66%, or 100%, respectively, of the metabolizable Lys from PBM was replaced with RP-Lys; SRC = sparing respiratory capacity; UCP = undigestible CP.

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